

System Architecture

Purpose

This document describes the structural organization of the Gildmark platform.

Rather than focusing on implementation technologies or individual services, it explains how the architectural philosophy is expressed through the organization of the system itself.

The architecture intentionally separates enduring operational concepts from implementation details, allowing the platform to evolve while preserving conceptual coherence.

The objective is not merely modularity.

It is the preservation of a consistent architectural model.

Architectural Overview

Gildmark is organized around operational capabilities rather than application modules.

The architecture is intentionally layered so that each level represents a different degree of abstraction and responsibility.

Higher layers define architectural intent.

Lower layers express that intent through progressively more concrete implementation.

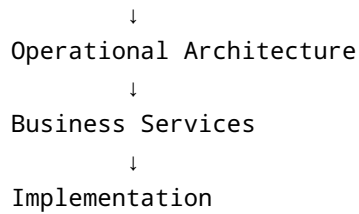
This organization allows architectural decisions to remain stable even as technologies, interfaces, and implementation strategies evolve.

Architectural Organization

At the highest level, the platform can be understood as six cooperating architectural layers.

```
graph TD; A[Architectural Philosophy] --> B[Design Principles]; B --> C[Architectural Decisions];
```

Architectural Philosophy
↓
Design Principles
↓
Architectural Decisions



Each layer derives naturally from the one above it.

No implementation decision should exist independently of the architectural reasoning that produced it.

Likewise, every architectural concept should ultimately be observable within the implementation.

Maintaining this traceability is fundamental to preserving architectural coherence.

Operational Architecture

The operational architecture organizes the platform around governed business capabilities rather than user interfaces or accounting functions.

Business activities are expressed as explicit operational lifecycles.

Lifecycle state governs behavior.

Operational events establish authoritative business history.

Accounting, reporting, documents, integrations, analytics, and user interfaces become consumers of operational truth rather than competing sources of truth.

The architecture therefore models the operation of the business before modeling the artifacts produced by the business.

Core Architectural Concepts

Several concepts appear repeatedly throughout the platform.

Operational Truth

The authoritative representation of business activity.

Operational truth serves as the foundation from which accounting, reporting, document generation, analytics, and integration are derived.

State-Driven Operations

Business behavior is governed through explicit lifecycle state.

Current state determines allowable operations, ensuring that business processes remain internally consistent and architecturally governed.

Lifecycle Governance

Every significant business process exists as a governed lifecycle with explicit transitions, invariants, and completion criteria.

Governance is embedded within the architecture rather than delegated to downstream validation.

Service-Oriented Operational Capabilities

Business capabilities are expressed through stable service contracts.

Services represent operational responsibilities rather than presentation workflows, allowing multiple consumers—including user interfaces, integrations, automation, and AI agents—to interact with the same governed capabilities.

Cross-Cutting Architectural Concerns

Several concerns intentionally span the entire architecture.

- Governance
- Traceability
- Provenance
- Operational consistency
- API stability
- Engineering documentation
- Architectural decision traceability

Rather than belonging to individual services, these concerns reinforce coherence across the platform as a whole.

Relationship to Implementation

Implementation technologies are intentionally treated as replaceable.

Programming language, persistence mechanisms, user interface technology, deployment strategy, and integration mechanisms exist to realize the architecture rather than define it.

This separation reflects one of the central themes of the platform:

Architecture should outlive implementation.

Closing Perspective

The structure of Gildmark is intentionally hierarchical.

Architectural philosophy informs design principles.

Design principles guide architectural decisions.

Architectural decisions organize operational capabilities.

Operational capabilities are implemented through services.

Implementation becomes the visible expression of an underlying conceptual model.

The architecture is therefore best understood not as a collection of modules, but as a coherent system of ideas expressed through software.